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SIMATS ENGINEERING

ASSESSMENT TOOL 3:

MCQ Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1596

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* **(BL3)**

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 15
Duration : 60 Minutes

Code of Conduct

I THIRU KUMARAN S / 192521049 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thiru Kumaran S

MCQ Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30%	Answers reflect strong and accurate concept mastery	Mostly accurate understanding	Partial understanding	Weak understanding
Accuracy of Responses	30%	Nearly all responses are correct	Most responses are correct	Some responses are correct	Few responses are correct
Application Awareness	15%	Shows strong recognition of practical cloud use cases	Reasonable recognition	Limited recognition	Weak recognition
Time Management	10%	Completes assessment efficiently	Completes on time	Needs extra time	Unable to finish well
Question Interpretation	15%	Interprets all items correctly	Misreads very few items	Misreads several items	Frequently misinterprets questions

Total Marks Awarded:

① A company's cloud storage system requires 5 TB of data replication across three region. what is the total storage required?

a) 5 TB b) 10 TB c) 15 TB d) 20 TB

Ans: c) 15 TB

② A cloud storage provider offers a 99.99 % uptime SLA. How much downtime can be expected per month.

a) ~4 minutes b) ~40 minutes c) ~4 hours d) ~24 hours

Ans: a) ~4 minutes

③ A cloud based load balancer distributes 60,000 request per second among 10 servers.

a) 5000 b) 6000 c) 7000 d) 10,000

Ans: 6000 (b)

④ A company's cloud network has 1 Gbps bandwidth and need to transfer 500 GB of data.

a) 1 hour b) 10 hours c) 1.1 hours d) 5.5 hours

(c) 1.1 hours

⑤ A cloud storage system experiences a 99.95 % availability rate.

a) ~4 hours b) ~8 hours c) ~20 hours d) ~ 40 hours

a) ~4 hours

⑥ A cloud Provider offer Pay-as-you-go pricing at $0.05 per gb for storage. If a company stores 10 TB, what is monthly cost?

a) $50 b) $500 c) $1000 d) $2000

b) $ 500

⑦ A cloud network has 10 gbps throughput but experi. 2% Packek loss?

a) 9.8 Gbps ✓ b) 9.6 Gbps c) 8.0 Gbps d) 7.5 Gbps

Ans: 9.8 Gbps

⑧ A user upload 250 mb of data per day to the cloud how much data will be uploaded. in one year?

a) 91.25 GB ✓ b) 85 GB c) 100 GB d) 120 GB

Ans: a) 91.25 GB

⑨ A cloud based video streaming service requires 5 Mbps bandwidth per user. If 100 user are streaming what is the total required bandwidth?

a) 500 mbps ✓ b) 250 mbps c) 200 mbps d) 1000 mbps

Ans: a) 500 mbps

⑩ A cloud Provider guarantees 99.95% uptime per year year how much downtime does this allow minute per year?

a) 262 minutes b) 365 minutes c) 525 minutes ✓ d) 150 minutes

c) 525 minutes

⑪ A virtualized data center has 200 Physical servers if each server run 40 virtual machines, what is total number of VMs?

a) 41000 b) 8,000 ✓ c) 6,000 d) 10,000

b) 8,000 tb

⑫ A Physical server has 128 gb of Ram and host 16 virtual machines each requiring 12 gb.

a) 4 gb b) 6 gb c) -8 gb ✓ d) 10 gb

c) -8 gb

(13) A virtual machines requires 4 v CPUs. If server has a total of 64 cores and an over commitment ratio of 2:1. how many VMs can be hosted?

a) 16 b) 32 c) 64 d) 128

b) 32

(14) A company migrates 50 on-Premise servers to the cloud, reducing consumpting by 60%. If the original Power usage 100 KW

a) 40 kw b) 50 kw c) 60 kw d) 70 kw

a) 40 Kw

(15) A cloud providers operator a virtualized data center with 100 physical server each with 64 v cpus. If the over commitment ratio is 4:1 how much total.

a) 6,400 b) 12,800 c) 25,600 d) 50,000

c) 25,600

(16) which cloud deployed model is used for organization that require both private and public?

a) Hybrid cloud b) Public cloud c) private cloud d) community cloud

a) Hybrid cloud

(17) which cloud security strategy ensures that sensitive remain private?

a) Encryption b) Load balancing c) Virtualization
d) Resource pooling.

a) Encryption.

18) If a hypervisor host 50 VMs and each VM uses 20 gb bandwidth ~~per~~ day calculate total bandwidth

a) 5000 gb b) 7000 gb c) 6000 gb d) 7500 gb

b) 7000 gb

19) A cloud system runs at 70% average CPU utilization If it has 32 cores at 2.5 GHz.

a) 56 GHz b) 58 GHz c) 64 GHz d) 70 GHz

a) 56 GHz

20) If the network latency in a virtualized system is 5 ms and processing time 10 ms, cal total round trip delay for 100 request

a) 1.5 sec b) 2 sec c) 1 sec d) 3 sec

a) 1.5 sec